Supplementary material for Adusumilli and Eckardt (2025)

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Equation (B.5) from Appendix B of Adusumilli and Eckardt (2025, "Temporal difference estimation of dynamic discrete choice models") describes the Riesz representer for the moment function $m(\cdot)$. In this note, we detail the steps leading to the derivation of this Riesz representer.

Observe that

$$m(a,x;\theta,h,g)=\sum_{\breve{a}\in\mathcal{A}}\left\{\mathbb{I}(a=\breve{a})-\pi(\breve{a},x;\theta,h,g)\right\}h(\breve{a},x).$$

For any $\tilde{h},\tilde{g}\in L^2(\mathbb{P})$, we can thus write

$$\begin{aligned}
&\mathbb{E}\left[m(a,x;\theta^*,\tilde{h},\tilde{g})-m(a,x;\theta^*,h,g)\right]\\
&=\mathbb{E}\left[\sum_{\breve{a}\in\mathcal{A}}\left\{\mathbb{I}(a=\breve{a})-\pi(\breve{a},x;\theta^*,h,g)\right\}\left\{\tilde{h}(\breve{a},x)-h(\breve{a},x)\right\}\right]\\
&\quad+\mathbb{E}\left[\sum_{\breve{a}\in\mathcal{A}}\left\{\pi(\breve{a},x;\theta^*,h,g)-\pi(\breve{a},x;\theta^*,\tilde{h},\tilde{g})\right\}h(\breve{a},x)\right]\\
&\quad+\mathbb{E}\left[\sum_{\breve{a}\in\mathcal{A}}\left\{\pi(\breve{a},x;\theta^*,h,g)-\pi(\breve{a},x;\theta^*,\tilde{h},\tilde{g})\right\}\left\{\tilde{h}(\breve{a},x)-h(\breve{a},x)\right\}\right]\\
&:=A_1+A_2+A_3.
\end{aligned}$$

Now, $A_1=0$ by the law of iterated expectations since $\mathbb{E}\left[\mathbb{I}(a=\breve{a})|x\right]=\pi(\breve{a},x;\theta^*,h,g)$. It thus remains to analyze A_2 and A_3. To this end, define $v(a,x):=h(a,x)\theta^*+g(a,x)$ and observe that

$$\pi(\breve{a},x;\theta^*,h,g)=\pi(\breve{a},x;\theta^*,v):=\frac{\exp\left\{v(\breve{a},x)\right\}}{\sum_{\bar{a}\in\mathcal{A}}\exp\left\{v(\bar{a},x)\right\}}.$$

Set $\boldsymbol{v}(x):=[v(0,x),v(1,x),\dots]^\intercal$ and take $\nabla_{\boldsymbol{v}}\pi(\breve{a},x;\theta^*,v)$ and $\nabla^2_{\boldsymbol{v}}\pi(\breve{a},x;\theta^*,v)$ to be the first and second Euclidean derivatives of $\pi(\breve{a},x;\theta^*,v)$ with respect to

the vector $\boldsymbol{v}(x)$. Now, a generic component of the vector $\nabla_{\boldsymbol{v}}\pi(\breve{a},x;\theta^*,v)$ is of the form

$$\partial_{v(\bar{a},x)}\pi(\breve{a},x;\theta^*,v) = \{\mathbb{I}(\bar{a}=\breve{a}) - \pi(\breve{a},x;\theta^*,h,g)\}\,\pi(\bar{a},x;\theta^*,h,g). \tag{1}$$

Clearly, $\sup_{\breve{a},\bar{a},x}\left|\partial_{v(\bar{a},x)}\pi(\breve{a},x;\theta^*,v)\right| < 1$ implying that

$$\left|\nabla_{\boldsymbol{v}}\pi(\breve{a},x;\theta^*,v)\right|_\infty < 1 \text{ uniformly over } \breve{a},x, \tag{2}$$

where $|\boldsymbol{b}|_\infty$ denotes the maximal element (in absolute value) of a vector $\boldsymbol{b}$. Taking a further derivative, we can similarly show

$$\left|\nabla^2_{\boldsymbol{v}}\pi(\breve{a},x;\theta^*,v)\right|_\infty < 1 \text{ uniformly over } \breve{a},x, \tag{3}$$

where $|B|_\infty$ denotes the maximal element of a matrix B.

We start with the term A_2. Denote $\tilde{v}(\breve{a},x) = \tilde{h}(\breve{a},x)\theta^* + \tilde{g}(\breve{a},x)$, $\tilde{\boldsymbol{v}}(x) := [\tilde{v}(0,x),\tilde{v}(1,x),\dots]^\intercal$, $\Delta v(\breve{a},x) := \tilde{v}(\breve{a},x) - v(\breve{a},x)$ and $\Delta\boldsymbol{v}(x) := \tilde{\boldsymbol{v}}(x) - \boldsymbol{v}(x)$. By a second order Taylor expansion and the mean value theorem,

$$\begin{aligned} A_2 = &\underbrace{-\mathbb{E}\left[\sum_{\breve{a}\in\mathcal{A}}\{\Delta\boldsymbol{v}(x)\}^\intercal\,\nabla_{\boldsymbol{v}}\pi(\breve{a},x;\theta^*,v)\big|_{\boldsymbol{v}=\boldsymbol{v}(x)}\cdot h(\breve{a},x)\right]}_{:=A_{21}} \\ &+\frac{1}{2}\underbrace{\mathbb{E}\left[\sum_{\breve{a}\in\mathcal{A}}\{\Delta\boldsymbol{v}(x)\}^\intercal\,\nabla^2_{\boldsymbol{v}}\pi(\breve{a},x;\theta^*,v)\big|_{\boldsymbol{v}=\dot{\boldsymbol{v}}}\{\Delta\boldsymbol{v}(x)\}\cdot h(\breve{a},x)\right]}_{:=A_{22}} \end{aligned}$$

for some $\dot{\boldsymbol{v}}$ between $\boldsymbol{v}(x)$ and $\tilde{\boldsymbol{v}}(x)$. In view of (1), the term A_{21} can be written as

$$\begin{aligned} A_{21} &= -\mathbb{E}\left[\sum_{\breve{a}\in\mathcal{A}}\left\{\sum_{\bar{a}\in\mathcal{A}}\{\mathbb{I}(\bar{a}=\breve{a}) - \pi(\breve{a},x;\theta^*,h,g)\}\,\pi(\bar{a},x;\theta^*,h,g)\cdot\Delta v(\bar{a},x)\right\}\cdot h(\breve{a},x)\right] \\ &= -\mathbb{E}\left[\sum_{\bar{a}\in\mathcal{A}}\pi(\bar{a},x;\theta^*,h,g)\left\{\sum_{\breve{a}\in\mathcal{A}}\{\mathbb{I}(\bar{a}=\breve{a}) - \pi(\breve{a},x;\theta^*,h,g)\}\,h(\breve{a},x)\right\}\Delta v(\bar{a},x)\right] \\ &= -\mathbb{E}\left[\left\{\sum_{\breve{a}\in\mathcal{A}}\{\mathbb{I}(a=\breve{a}) - \pi(\breve{a},x;\theta^*,h,g)\}\,h(\breve{a},x)\right\}\Delta v(a,x)\right] = \langle\psi(\cdot,\cdot;\theta^*,h,g),\Delta v\rangle\,, \end{aligned}$$

where the third equality follows from the law of iterated expectations since $\pi(\bar{a},x;\theta^*,h,g)$ is the true conditional choice probability. Noting the definitions of $\psi_h(\cdot),\psi_g(\cdot)$ and $\Delta v(\cdot)$, we conclude

$$A_{21} = \left\langle\psi_h(\cdot,\cdot;\theta^*,h,g),\tilde{h}-h\right\rangle + \langle\psi_g(\cdot,\cdot;\theta^*,h,g),\tilde{g}-g\rangle\,. \tag{4}$$

We now turn to the term A_{22}. Under the assumptions of compact support for x and the continuity of h,g, there exists $C<\infty$ and $\delta>0$ such that

$\inf_{a,x} \pi(a, x; \theta^*, h, g) > \delta > 0$ and $|h|_\infty < C$. Consequently, in view of (3), we can bound A_{22} as

$$\begin{aligned}
|A_{22}| &\leq C|\mathcal{A}|^2 \mathbb{E}\left[\sum_{\breve{a}} \{\Delta v(\breve{a}, x)\}^2\right] \\
&\leq C|\mathcal{A}|^2 \delta^{-2} \mathbb{E}\left[\sum_{\breve{a}} \pi(\breve{a}, x; \theta^*, h, g) \{\Delta v(\breve{a}, x)\}^2\right] \\
&= C|\mathcal{A}|^2 \delta^{-2} \|\Delta v(\breve{a}, x)\|_2^2 \ \leq 2C|\mathcal{A}|^2 \delta^{-2} \left\{\theta^{*2} \left\|\tilde{h} - h\right\|_2^2 + \|\tilde{g} - g\|_2^2\right\}. \quad (5)
\end{aligned}$$

It remains to bound A_3. By (2),

$$\begin{aligned}
|A_3| &\leq \mathbb{E}\left[\left\{\sum_{\bar{a} \in \mathcal{A}} |\Delta v(\bar{a}, x)|\right\} \left\{\sum_{\breve{a} \in \mathcal{A}} \left|\tilde{h}(\breve{a}, x) - h(\breve{a}, x)\right|\right\}\right] \\
&\leq |\mathcal{A}| \mathbb{E}^{1/2}\left[\sum_{\bar{a} \in \mathcal{A}} |\Delta v(\bar{a}, x)|^2\right] \mathbb{E}^{1/2}\left[\sum_{\breve{a} \in \mathcal{A}} \left|\tilde{h}(\breve{a}, x) - h(\breve{a}, x)\right|^2\right].
\end{aligned}$$

By a similar argument as in the bound for $|A_{22}|$ above,

$$\mathbb{E}^{1/2}\left[\sum_{\bar{a} \in \mathcal{A}} |\Delta v(\bar{a}, x)|^2\right] \leq \delta^{-1} \left\{\theta^* \left\|\tilde{h} - h\right\|_2 + \|\tilde{g} - g\|_2\right\}.$$

Also in a similar vein,

$$\mathbb{E}^{1/2}\left[\sum_{\breve{a} \in \mathcal{A}} \left|\tilde{h}(\breve{a}, x) - h(\breve{a}, x)\right|^2\right] \leq \delta^{-1} \left\|\tilde{h} - h\right\|_2.$$

We thus conclude

$$|A_3| \lesssim \left\{\left\|\tilde{h} - h\right\|_2 + \|\tilde{g} - g\|_2\right\}^2. \quad (6)$$

Equation (B.5) in Adusumilli and Eckardt (2025) then follows by combining (4) - (6).